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Personalised healthcare model for monitoring and prediction of airpollution: machine learning approach

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ABSTRACT

The drastic increase in atmospheric pollutants has resulted in the prevalence of hazardous diseases like Asthma, Ischaemic heart disease, and Pulmonary disease around the world. IoT technology has the capability to acquire and monitor air quality parameters in the ambient environment of an individual. Inspired from these aspects, this paper proposes an IoT-based automated framework for monitoring and predicting air quality parameters like benzene using machine learning technology. Specifically, this study incorporates an Adaptive Neuro-Fuzzy Inference System (ANFIS) technique to predict the air quality in the form of Level of Pollutant (LoP) and modified Air Quality Index (m-AQI). Moreover, for the optimisation of ANFIS technique, Differential Evolution (DE)-inspired algorithm has been proposed for overall enhancement of system accuracy. In order to validate the proposed model, numerous experimental simulations were performed over four challenging datasets and results were compared with several state-of-the-art models. Comparative analysis shows improved statistical values for the presented model in terms of Accuracy (97.3%), Coefficient of Determination (94.01%), and Root Mean Square Error (2.4%). In addition to these, enhanced values of performance estimators like Reliability (92.36%) and Stability (76.00%) were estimated for depicting overall efficiency of the proposed system.

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KEYWORDS
Internet of Things (IoT); modified-air quality index; adaptive neuro-fuzzy inference system; differential evolution

Introduction

Air contamination is one of the significant causes of severe health diseases like asthma, bronchitis, and chronic obstructive pulmonary disease (COPD), which can lead to permanent health loss, and death (Pannu et al. (2017). In fact, according to a survey report by Global Burden of Disease in 2017, air pollution has become one of the top 10 dangers confronted by human beings.[1] Air contamination happens with the presence of the inordinate amount of harmful substances. For gaseous substances, these include an excessive amount of nitrogen dioxide (NO_2), benzene (C_6H_6), carbon dioxide (CO_2), methane (CH_4), ozone (O_3), carbon monoxide (CO), nitric oxide (NO), sulphur dioxide (SO_2), particulate matter (PM) and other particles such as dust, pollen, smoke, and liquid droplets. An uncontrolled presence of air contamination has been a major cause of life-loss in the recent past. Factually, a survey conducted by World Health Organisation (WHO) showed that the open-air toxicity caused nearly 4.2 million premature deaths globally in 2016.[2] More than 90% of the deaths have been seen from developing and underdeveloped nations. Another survey conducted by the American Lung Association (ALA) reported that more than 40% of United States (US) population is in danger of severe health diseases like cancer, coronary illness, stroke, and asthma due to presence of excessive air pollutants like benzene (C_6H_6) and sulphur dioxide (SO_2).[3] Some other diseases include chronic

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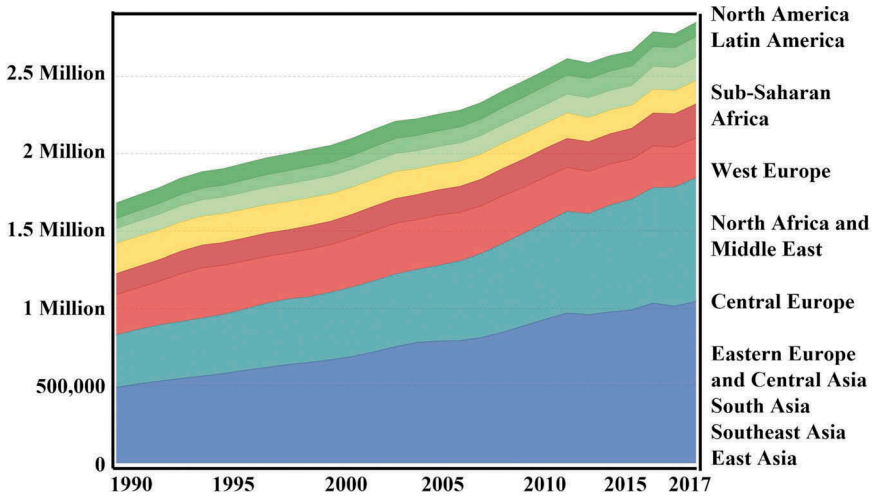

Figure 1. Increase in Global Mortality Rate due to Air Pollution.

obstructive pulmonary disease, lung cancer, ischaemic heart disease, and severe stroke (Nel, 2005). With the continuous global industrialisation leading to an increase in air pollution, disease vulnerability is increasing at enormous rate resulting in temporary or permanent health loss and even deaths to common people. The key sources of air contamination are broadly classified into four groups (i) transportation sources, (ii) static sources, (iii) regional sources and (iv) natural sources. Moreover, pollution may arise due to the waste material generated by different sources like oil refineries, commercial organisations, and factories. These generate a huge amount of air pollutants while processing raw material or manufacturing of goods. One of the other indispensable cause of air pollution is due to particular areas like agriculture areas, metro cities, or wood-burning fireplaces. Nevertheless, natural sources like wind-blown dust, volcanoes, and wildfire are also sometimes significantly considered as the main source of air pollution. The number of contaminants a city or nation discharge can cause an increase in mortality rate or serious medical issues. In fact, Figure 1 shows the increase in regional mortality rate over the past few years.[4] Thus, air monitoring has become an indispensable requirement to monitor the rate of air contamination for preventing human health loss.

Research field

Advancements in Information and Communication Technology (ICT) have formed a bedrock for developing innovative solutions in various industrial sectors like healthcare, logistics, transportation, and agriculture (Bhatia et al., 2019; Gubbi et al., 2013). With the Internet of Things (IoT) acting as one of the major ICT drivers for technological progress, the futuristic industries are led on the path of automation and decentralised intelligence. In addition to this, by developments of intelligent and inter-networked IoT devices, the interaction of real and cyber world has proven to be a considerable breakthrough for automating numerous industrial services. Development of numerous IoT sensors for acquiring ubiquitous air pollutants like benzene (C_6H_6), and sulphur dioxide (SO_2) has enabled real-time air quality assessment. In addition to this, the harmful impact on human health can be effectively monitored due to the presence of vulnerable air components in real-time (Manogaran et al., 2018). Moreover, with the utilisation of artificial intelligent techniques over fog computing technology, services like data perception, analysis, and real-time prediction can be obtained with high efficacy (Bhatia & Sood, 2018). Furthermore, the incorporation of visualisation techniques has resulted in effective presentation of decision-making services to users in real-time (Chen et al., 2018).

Major contribution

Conventionally, Air Quality Index (AQI) has been considered as one of the indicators of air quality as it ranges from 0 to 500. The estimation of air contamination relies upon the AQI level of a region. A high value prompts an expansion in the contamination range and well-being concern (Raipure, 2014). More precisely, an estimation of AQI of 40 is considered as acceptable air quality dimension with minimal potential to influence general well-being, while the value of AQI more than 300 demonstrates the hazardous air quality. Air toxins such as CO2, CH4, CO, NO2, O3, and PM are considered as six significant gases of air contamination (Raipure, 2014). However, such estimation measure is an approximation measure for determining the presence of air pollutants as it is based on atmospheric data. In other words, real-time air contamination and statistical evaluation of regional air pollution are not taken into consideration. In order to provide a real-time estimation of regional air pollutants such as benzene, this research incorporates machine learning technique termed as ANFIS for time-sensitive determination of air contamination in a smart air quality monitoring system. Specifically, the objectives of the proposed monitoring system are as follows:

(1) Utilise IoT-Fog computing for acquiring ambient air quality parameters in real-time for detection of air pollutants.
(2) Classify acquired IoT data segments based on the Level of Pollutant (LoP) measure for probabilistic estimation in terms of modified-Air Quality Index (m-AQI) for vulnerable air quality parameters.
(3) Estimate and predict the relationship between the contaminated features of ambient air and spatio-temporal aspects through an adaptive neuro-fuzzy inference system (ANFIS) in an optimal manner.
(4) ANFIS optimisation using Differential Evolution (DE) Method for enhancing overall system performance.
(5) Visualise the predictive results using Self-Organised Mapping (SOM) technique for provisioning real-times services.
(6) Evaluation of the proposed system efficiency in terms of Accuracy, Coefficient of Determination, Reliability, and Stability.

Paper Organisation

The rest of the paper is organised into various sections. Section 2 provides a brief overview of numerous contributions in the field of air quality monitoring and data analysation techniques. The proposed model of smart-air quality monitoring and prediction has been provided in Section 3. Section 4 provides a detailed experimental simulation for validation purposes. Section 5 presents a comparative analysis with six state-of-the-art air quality monitoring systems. Finally, Section 6 concludes the paper with some important future research directions.

Related works

This section discusses some of the important contributions in the field of air quality monitoring systems that have been put forth by researchers around the world. Specifically, two subsections have been formed reviewing significant contributions in IoT-inspired air quality monitoring system and how artificial intelligence techniques have been incorporated by researchers for provisioning effective health-oriented services with predictive decision-making.

IoT-inspired air quality management systems

IoT technology has been adopted by numerous researchers for providing frameworks for monitoring air-quality in real-time. Barthwal et al. (2019) collected and analysed air quality information around

the regions of New Delhi, India by utilising IoT technology. The presented system was able to acquire ambient air data on an hourly basis, which is further used to forecast exceedance probabilities of air quality. For this, extreme value distribution has been used to compare with vulnerable events. Based on the simulation implementation, the authors deduced that the presented system was highly effective in monitoring air quality and provided necessary indications at an early stage. Biswal et al. (2019) have utilised IoT technology to ascertain air quality based on the mathematical formulation in indoor environments. Authors have incorporated cloud computing technology for storing and manipulation of data acquired by numerous IoT devices embedded in the room environment. For simulation purposes, ThingSpeak was used for determining the performance of the overall system. Based on the experimental simulations, effective results were registered in different experimental scenarios. Lai et al. (2019) proposed a low-cost air quality monitoring and real-time prediction system using IoT technology and fog computing. Specifically, the authors utilised Raspberry Pi as fog node and processed Kalman filters algorithm for enhancing the performance of IoT sensors. Moreover, the proposed system was able to acquire better performance in the acquisition of SO2, NO2 and PM2.5. A similar model was presented by Guo et al. (2019). In the proposed research, authors presented a comprehensive case study of IoT-cloud participatory sensing where a user sends a query to the cloud considering the air quality of a location at a specific time. IoT devices were able to acquire ubiquitous air quality data and transmit the data to cloud storage for further analysis. Real-time user service delivery was one of the major objectives of the proposed system. Experimental simulations were able to register enhanced performances in comparison to other state-of-the-art techniques. Hu et al. (2019) presented a real-time, fine-grained, and power-efficient air quality monitoring system-based sensors embedded in aerial and ground sensing. The architecture of the presented system was comprised of different layers for sensing, transmission, processing, and presentation. In addition to this, the deployment strategy and power control are two major challenges addressed by authors in the presented study. The proposed model has been successfully deployed in Peking University and Xidian University in 2018 and was able to acquire 100,000 values over a specific time period. Thosar and Nathi (2019) proposed an effective air quality measurement system for tracking air quality in the regional atmosphere. Specifically, the authors incorporated internet-enabled IoT sensors for an end to end data transmission. The parameters sensed by the system were transmitted server through an interconnected cellular network system. Communication between various devices and server is deployed using Message Queuing Telemetry Transportation (MQTT) protocol for enhanced system performance. Experimental simulation shows better results in comparison to other models.

Artificial intelligence techniques for predictive analysis

Over the past few years, numerous strategies have been proposed by researchers around the world for air quality prediction and estimation of air contamination. The time series and regression-based methods are the most widely used methods for scale prediction. Kleine Deters et al. (2017) have utilised the machine learning-based statistical models to predict the scale of pollution from the meteorological data by comparing the level of Particulate Matter (PM) in the air. Gautam et al. (2019) analysed the grid-based environment and also explained the effects of mesoscale type wind with respect to the toxins discharged from a slit source. The examination infers that the wind exasperates the groupings of contamination. Brandt et al. (2001) developed an internationally acknowledged model named THOR for air quality applications. The utilisation of this model has been done in order to anticipate the climate and contamination range for 3 days. Park et al. (2018) proposed an air monitoring framework to predict the concentration of PM in an indoor environment by utilising an Artificial Neural Network (ANN) system. Basically, the authors monitored the air quality of subway stations. Authors calculated the correlation between the prediction performance and the depth of a station. By calculating the correlation, the enhanced performance was registered for the variable depth of the station. Li et al. (2018) estimated the concentration of pollutants in the air by using

Support Vector Regression (SVR) and Stepwise Regression (SWR) methods. They predicted PM concentration from the real-time readings acquired by the air quality monitoring-based sensor network. The calculated results justified the concentration prediction accuracy of the predictable methods on the data of nearby stations. Karatzas et al. (2018) proposed a combined approach of Linear Regression, Artificial Neural Network (ANN), and Random Forest to predict the concentration of the contaminants in the environment. The results demonstrated the better performance of the presented model for prediction of PM concentration. Franceschi et al. (2018) used data mining techniques for prediction of the concentration of PM10 in the atmosphere. They analysed the behaviour of the data by detecting the influential value. Furthermore, k-means clustering methods along with ANN were designed to analyse the range of PM10 and PM2.5 in the atmosphere. Principal Component Analysis (PCA) results are transferred to ANN for prediction purposes and k-means clustering was used for grouping the results for further analysis. Kumar and Kumar (2018) utilised the random forest method in combination with DE to predict the scale of air pollution. The proposed approach gains higher prediction performance and outperforms other state-of-art models. Seven pollutants such as NO_2, C_6H_6, SO_2, PM2.5, PM10 and O_3 were considered for prediction of air pollutant concentration.

Proposed methodology

Figure 2 presents a conceptual overview of the proposed methodology for smart air quality monitoring and prediction system. Specifically, the presented model comprises five important layers, namely, Data Acquisition (DA) layer, Data Classification (DC) layer, Data Mining and Extraction (DME) layer, Prediction Decision Making (PDM) layer, and Visualisation (VL) layer for air quality prediction. In

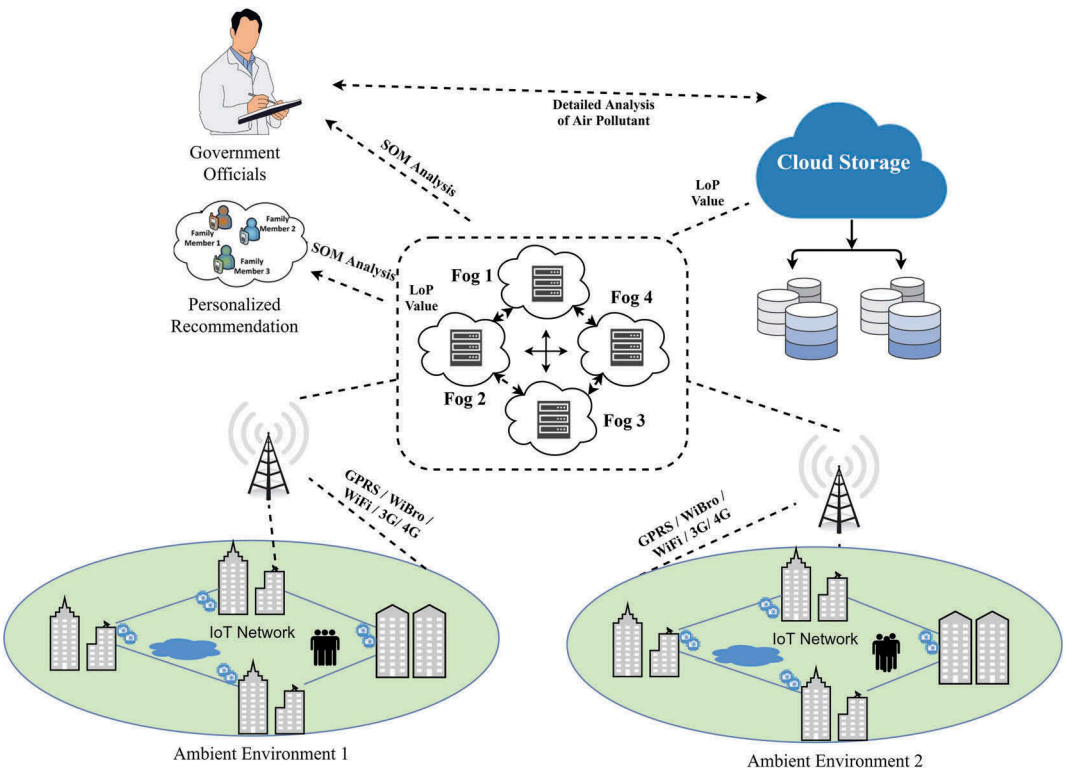

Figure 2. Conceptual Framework of Proposed Model.

the initial layer, data is acquired ubiquitously using numerous IoT sensors, geographically displaced in the ambient environment over building, poles, and roof-tops. Acquired data comprising numerous air parameters including Benzene (C6H6), SO2, PM(2.5), and PM(10) are transmitted to the remotely connected fog computing device. DC layer enables categorisation of data segments into different data sets based on its harmful impact, which is extracted in real-time using spatio-temporal data extraction technique by DME layer. Based on data sets, the Adaptive Neuro-Fuzzy Inference System (ANFIS) backed by Differential Evolution (DE) mechanism is proposed to effectively predict the presence of air pollutants over geographical regions. Finally, VL layer is used to visualise the level of pollution using colour coding technique for early warning and alert generation. The layered architecture of the proposed model is shown in Figure 3. The detailed functionality of each layer has been provided ahead.

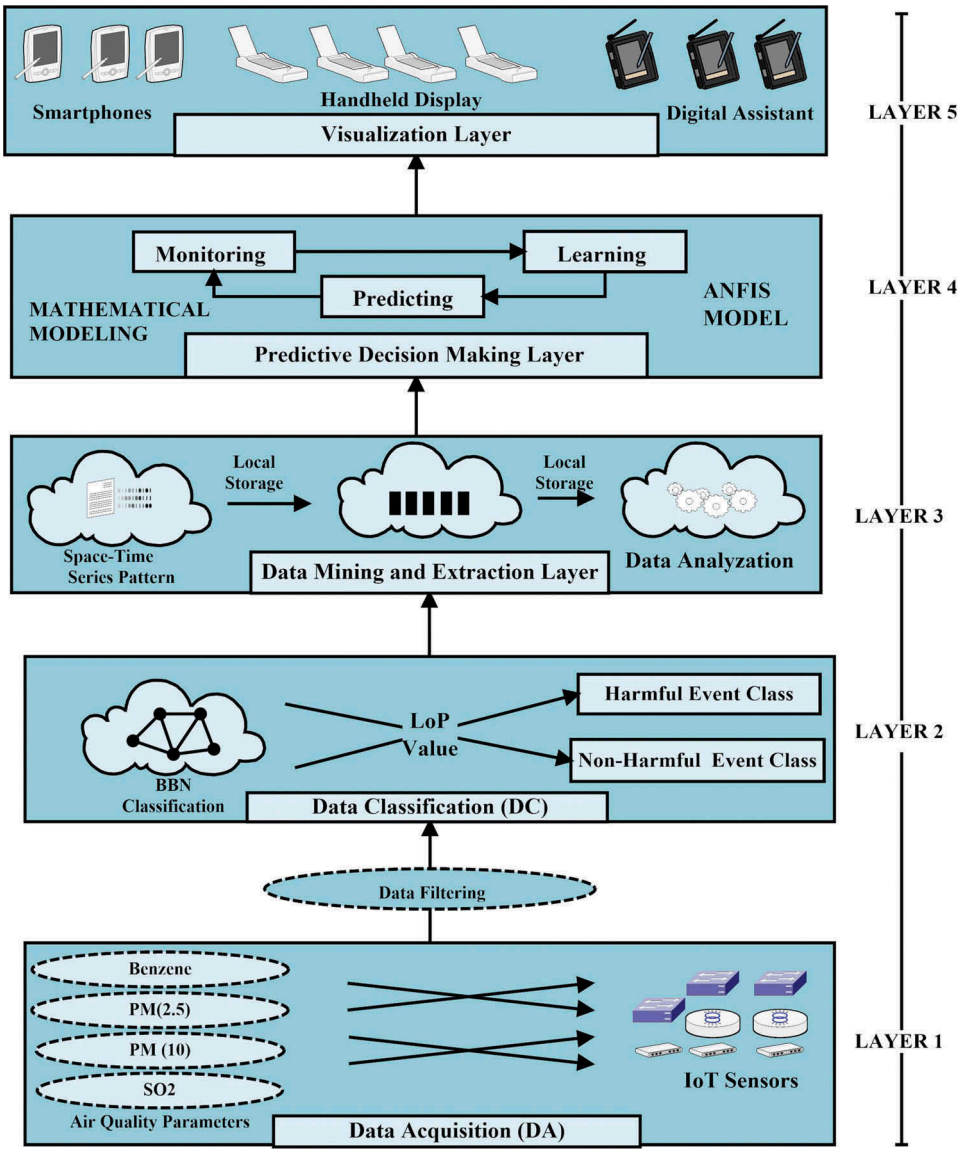

Figure 3. Proposed Architecture for Real-time Air Quality Monitoring and Prediction.

Data acquisition (DA) layer

Data Acquisition is the initial layer in the proposed framework of spatio-temporal air quality analysis. As mentioned earlier, numerous IoT sensors, actuators, and wireless devices are used over geographically distributed buildings, street poles, and rooftops to acquire real-time data. Geographical coordinates are acquired by GPS system embedded on these devices. Some of the commercially available devices include Sparkfun, MQ135, Turbidity Sensor, and Air Contaminator Sensor. These devices are cheap devices and have the capability of data transmission to the remote location using numerous wireless communication technology like Bluetooth, Zigbee, and WiFi. Moreover, IoT devices are equipped with an internal clock structure for provisioning real-time services. On the contrary, acquiring data from numerous devices may result in temporal inconsistencies. In fact, Clarke et al. (2014) have categorised sensors based on clock structure as depicted in Table 1. For temporal homogeneity, a remote device (gateway) is programmed to synchronise the data segments based on a universal clock structure. Data acquired from IoT devices is stored in the local computing device which is further connected to cloud storage for detailed analysis. In the current research, Raspberry Pi-based computation has been performed to provide real-time air quality monitoring and prediction. In addition to this, cloud storage is used for real-time visualisation. Furthermore, researchers around the world have provided numerous feature abstraction techniques and mechanisms that can be efficiently utilised for heterogeneous data segments like numerical, textual, and imagery. Li and Guo (2011) have proposed a symbolic aggregate approximation (SAX) Algorithm for abstracting features like temporal data and textual data. Similarly, Xu et al. (2017) have proposed procedural graph model for RFID data (numeric and non-numeric). For security perspective, protocols like Secure Socket Layer (SSL) and Elliptical Curve Cryptography (ECC) are used for provisioning data security during transmission (Bhatia & Sood, 2019).

Data classification (DC) layer

Data Classification is the second layer of the proposed model intended to categorise data segments for air quality assessment. The proposed research incorporates numerous data segments for the presence of Benzene (C_6H_6), SO_2, $PM_{(2.5)}$, and $PM_{(10)}$. As mentioned earlier, air pollution can be caused by a large number of contaminants. In order to confine this diversity, the current research focusses on analysing these four parameters for air quality. In addition to this, each of the parameters is assigned a priority for determination of air pollution with respect to the ambient environment. For instance, benzene pollutant is assigned higher priority in regions which are prone to benzene emission. This is due to the fact that these pollutants are the most adverse contaminants that are responsible for health diseases like Asthma, COPD, Blood cancer leading to deaths. Henceforth, each of these parameters is essential to determine the overall quality of air in real-time. The details of these harmful pollutants are discussed ahead.

1. *Benzene (C_6H_6)*
   Benzene is a colourless substance available in both gaseous and liquid form. It evaporates at a high rate and is highly inflammable. Exposure to benzene components in the air (vehicles emission, tobacco smoke, glues, paint, and detergents) is harmful to human health. Inhalation of extreme levels of benzene results in damage to blood-forming organs and can lead to life

Table 1. Clock Synchronisation for IoT Sensors.

S.No.	Category	Description
(1)	No Clock	Sensors do not have internal clock
(2)	Non-Synch	Sensors do have internal clock but is not synchronised
(3)	Absolute Time	Sensors sense data depending upon absolute time
(4)	Relative Time	Sensors sense data relative to other device.

loss if no treatment is considered. In addition to this, health diseases like asthma, and lung cancer are other serious health effects for inhaling benzene components from the air.

2. *Sulphur Dioxide (SO_2)*
   Sulphur dioxide is an invisible gas having a nasty and sharp smell. It affects human health severely as it irritates nodes, throat, and airways to cause coughing, shortening of breath and wheezing. In some cases, chest pains are caused due to the breathing of this gas. Its effects are felt quickly as it takes nearly 10 to 15 min to affect human health.
3. *Particulate Matter ($PM_{2.5}$)*
   Fine Particulate Matter ($PM_{2.5}$) is an air pollutant that can affect people's health severely. These are the particles that reduce the visibility in the air and cause haziness when present at extreme levels. These particles can travel deeply in respiratory level when inhaled and can damage lungs significantly. Moreover, prolonged exposure to these particles can result in short-sightedness, throat and lung infection. In most of the cases, these particles are the cause of asthma and heart diseases. Conspicuously, the damage due to such pollutants can be enormous, especially in old patients with minimised health immunity.
4. *Particulate Matter* (PM_{10})
   Particulate Matter (PM_{10}) is similar to Particulate Matter ($PM_{2.5}$) except for the difference in their sizes. These pollutants are also considered as one of the most significant parameters for assessment of the AQI. Presence of a large quantity of PM_{10} can result in severe health problems like asthma, infection, and itching.

Based on these different air pollutants, two categories for monitoring air quality are defined. These classes include harmful event and non-harmful event class. Harmful event class comprises air quality values for pollutants that are vulnerable to human health. Similarly, for pollutants which are a not present in abundance are confined to non-harmful event class. In the current study, Bayesian Probabilistic Models (BPM) have been used for classification of datasets into one of the two classes because of its high efficiency. BPM is a probabilistic modelling technique for predicting the posterior probabilities based on prior probabilities. In order to analyse the harmful class events, it is necessary to quantify air-based harmful pollutants into a unifying factor. For this purpose, the current research utilises the incorporation of a quantification measure termed as Level of Pollutant (LoP) for real-time air quality assessment.

Definition 1: Level of Pollutant (LoP) *Level of Pollutant (LoP) is defined as a certainty measure to determine the presence of an air pollutant value at a given time window Δt and in the vicinity Δs. In other words, LoP provides a quantification of the air quality in a time window Δt for the presence of harmful air particles that can affect human health in an adverse manner in a given geographical space of Δs.*

Definition 1 provides a comprehensive overview of provisioning an effective measure to analyse air quality in real-time over a geographical space of Δs.

Mathematical analysation: temporal aspect

BPM is a probabilistic model used for estimating a class for a particular data segment. Specifically, let $P(Q_T/x)$ and $P(Q_F/x)$ denote the 'True' and 'False' respective probability of air quality under the observation attribute of $x > \lambda$ (prefixed threshold) at time T, and w be the associated weight for attribute x. The weighted BPM model defines the weighted prior probability $P_w(x > \lambda/Q_T)$ as follows:

$$P_w(x/Q_T) = w(x) * P(x/Q_T)$$

The weighted posterior probability $P_w(Q_T/x)$ is a probability function, given the definition of the weighted prior probability $P_w(x/Q_T)$. Accordingly, the probability $P_w(Q_T/x)$ can be inferred as follows:

$$P_w(Q_T/x) = P_w(x/Q_T) * P(Q_T)P(x)$$
$$P_w(Q_T/x) = w(x) * P(x/Q_T)* P(Q_T)\ P(x), \text{ where}$$
$$P(x) = P_w(x/Q_T)P(Q_T) + P(x/Q_F)P(Q_F)$$
$$P(x) = w(x)P(x/Q_T)P(Q_T) + P(x/Q_F)P(Q_F)$$

Then, after accounting for n attributes for a given dataset, the vulnerability score or LoP value is evaluated as

$$\text{LoP} = \frac{1}{n}\sum_x P_w(Q_T/x)$$

Data mining and extraction (DME) layer

Data mining is one of the important layers in the proposed model. The main objective of this layer is to extract data from the storage repository for analysation. In the current research, the task of data mining is performed using spatio-temporal mining technique. Spatio-Temporal Mining is an effective data mining technique capable of extracting information in the form of sequential pattern depending on the centralised clock and GPS coordinates. This study defines a spatio-temporal granulation technique for retrieving data in a given time window Δt and space window of Δs.

Definition 2: Spatio-Temporal Granulation $_{\Delta t,\Delta s}$ *Spatio-Temporal Granulation ($STG_{\Delta t,\Delta s}$) is defined as a sequential pattern for the data stored in the repository associated with the spatio-temporal instance t_1, t_2, and so on, and s_1, s_2, and so on. Mathematically, Spatio-Temporal granulation for data instances d_1, d_2 and so on can be represented in a given time and space window Δt and Δs as $STG_{\Delta t,\Delta s} = [(d_1,t_1,s_1)\ (d_2,t_2,s_2)\ \ldots .(d_n,t_n,s_n)]$.*

Definition 2 provides a comprehensive accumulation of air quality parameters from the storage repository. Figure 4 provides a brief overview of the extraction of air quality parameters based on the clock window Δt and geographical space window Δs. For analysis of the air quality parameters based on LoP value, a probabilistic parameter is defined in terms of modified-Air Quality Index (m-AQI). m-AQI is different from the conventionally adopted quality index (AQI) in the way that in the current study it is estimated based on the respective LoP values of air pollutant.

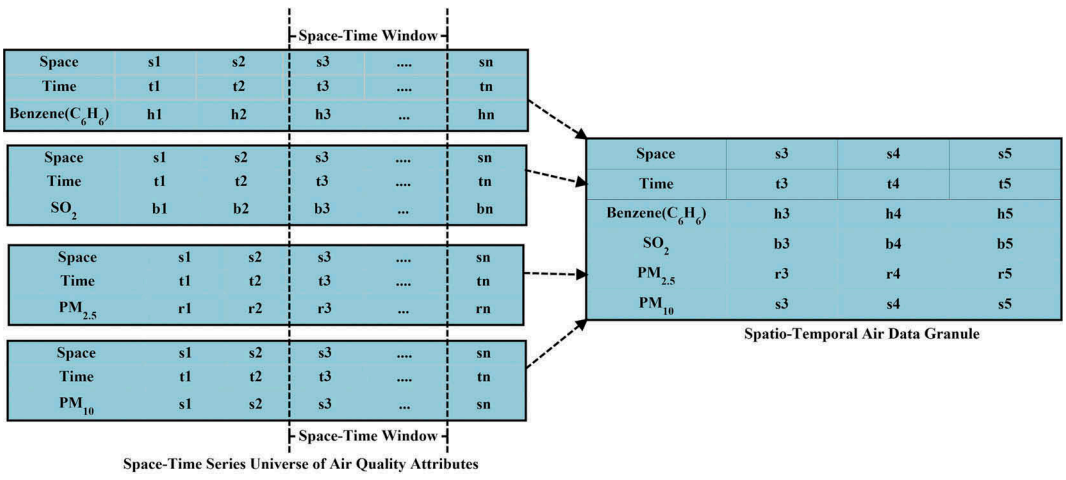

Figure 4. Spatio-Temporal Granulation for Air Quality Monitoring.

Table 2. Modified-Air Quality Assessment for Harmful parameters.

Algorithm 1: modified-Air Quality Assessment by Prioritising Pollutants
Step 1: Input IoT data values for n air quality parameters and associated LoP values. α, β, γ, δ are the associated priority weights.
Step 2: Initialise m-AQI$_{\Delta}$t = 0.
Step 3: Compare LoP value of air quality parameter 1 with prefixed threshold value.
Step 4: If LoP$_1$ > λ_1, Then Add α *LoP$_1$ to m-AQI.
Step 5:Compare LoP value of air quality parameter 2 with prefixed threshold value.
Step 6: If LoP$_2$ > λ_2, Then Add β *LoP$_2$ to m-AQI
Do for n parameters
Step 7: Compare LoP value of air quality parameter n with prefixed threshold value.
Step 8: If LoP$_n$ > λ_n, Then Add β *LoP$_n$ to m-AQI
Step 9: Cumulative m-AQI = α *LoP$_1$ +β *LoP$_2$ +γ *LoP$_3$+ ……..δ *LoP$_n$

Definition 3: modified-Air Quality Index(m-AQI) *m-AQI is defined as a probabilistic measured for aggregating the vulnerable impact of air pollutants in given time window Δt. In other words, m-AQI provides a comprehensive indicator for air quality of the ambient environment in real-time and space.*

Definition 3 provides a novel m-AQI measure for estimating the probability of air pollutant in given space-time window. Moreover, it presents an effective quantified parameter for pollution assessment using IoT technology. In addition to this, the current research focuses on acquiring the LoP value for specific air pollutant (for instance Benzene) in the ambient environment. The detailed steps for evaluating the harmful event class with prioritisation of air quality parameters are shown in Table 2. In Algorithm 1, α, β, γ, δ denotes the associated weights having the value in the range of (0,1). For instance for prediction of air quality with respect to benzene parameter, the associated weight will be 1 while the rest of the weights will be 0.

Predictive decision making (PDM) layer

Adaptive Neuro-Fuzzy Inference System (ANFIS) model is applied in numerous domains for predictive decision-making by researchers around the world (Nauck et al., 1997). ANFIS is capable of extracting rules from the sampled input data by the adapting principles of Neural Network (Pannu et al., 2018). The ANFIS system incorporates a training and learning algorithm for creating fuzzy reasoning. The ANFIS makes the system more capable of self-adaptive and self-organised by tuning rules automatically. In the current research, each air quality parameter is assessed by the proposed ANFIS model within a given space-time window. For instance, the data values for the presence of benzene in the ambient environment in a given space-time window are fed to the ANFIS for which its predictability value is estimated. Figure 5 represents the architecture of ANFIS. Proposed ANFIS comprises five important layers in which each layer includes several nodes represented with the help of the node function and hidden layers. Moreover, each of the layers is assigned a hidden weight for normalisation purposes. The detailed mathematical analysation of ANFIS layers is discussed ahead.

(a) Fuzzification (Layer 1): The initial layer of the ANFIS model is responsible for the task of fuzzification. The membership grade related to each fuzzy set is produced by each node of this layer. In this layer, every node is adaptive in nature and represented as:

$$N_i^1 = \mu H_i(x) \tag{1}$$

where N_i^1 is the generalised gaussian membership function, x is considered as the input value provided to the node i, and H_i is considered as the linguistic label which is further linked with the node function (Reddy & Raju, 2009). In the current scenario, spatio-temporal data values for different air quality parameter instances are provided to the initial fuzzification layer.

(b) Product rule (Layer 2): The nodes of the second layer transfer the input signal to the next layer by performing element-wise product operation represented as follows:

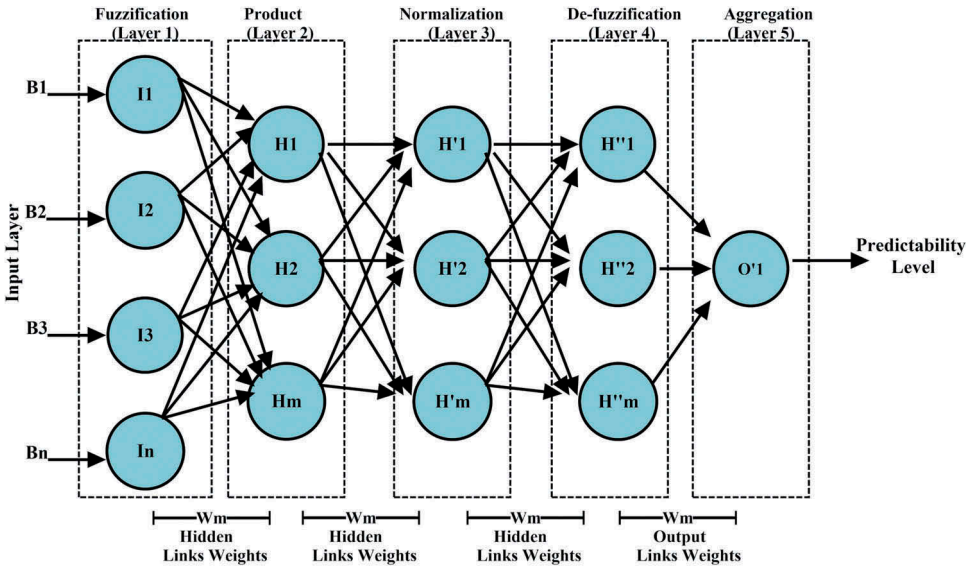

Figure 5. ANFIS Approach for Benzene Prediction.

$$N_i^2 = \mu A_i(x) \times \mu B_i(y),\ i = 1, 2 \tag{2}$$

where Ai(x) and Bi(y) denotes the nodes in layer 2.

(c) Normalisation (Layer 3): Every single node of this layer is responsible for calculating the ratio of single firing strength rule to the sum of all firing strength rules as represented in the Equation (3). w_i' represents the firing strength which is normalised as follows:

$$N_i^3 = w_i' = \frac{w_i}{w_1 + w_2} for\ i = 1, 2 \tag{3}$$

(d) De-fuzzification (Layer 4): The responsibility of this layer is to evaluate the contribution of the ith rule to the final output. The parameters c_i, d_i, r_i are known as consequent optimised variables. This layer performs the function of de-fuzzification as follows:

$$N_i^4 = w'if_i = w'i(c_i x + d_i y + r_i)\ for\ i = 1, 2 \tag{4}$$

(e) Output generation (Layer 5): This layer is termed as output layer and is responsible for the calculation of sum of all outputs which are originating from all nodes of de-fuzzification layer and computes the final output Ni5 as represented in Equation (5):

$$N_i^5 = \sum w_i' f_i = \frac{\sum_i w_i f_i}{\sum_i w_i} \tag{5}$$

In the current scenario, this output corresponds to the predicted value for the given space-time window and overall m-AQI value is estimated. The effectiveness of the presented technique lies in the fact that the overall model comprises efficient mathematical formulation and therefore effective services can be provided in real-time. Moreover, the proposed ANFIS model is backed by Differential Evolution (DE) technique for optimisation over accumulated instances of air pollutants. The details of DE-based optimisation are provided ahead.

Differential evolution-inspired optimisation

Several optimisation algorithms are proposed by researchers to optimise the performance of the models, such as Genetic Algorithm (GA), Differential Evolutionary (DE) and Particle Swarm Optimisation (PSO). In the proposed air quality prediction framework, DE-based evolutionary algorithm is employed to normalise the model's parameters (Storn & Price, 1997). The working of all these algorithms is based on stochastic parallel algorithms which are the part of swarm intelligence. All these algorithms are following the common procedure, such as crossover, mutation and selection. DE follows a different process of mutation and selection as compared to the common basic procedure (El Ela et al., 2010; Sha & Hsu, 2008). Specifically, DE presents new offspring by the formation of the trailing vector of each individual parent of the population. This is further improved by the fundamental operators of mutation, crossover and selection. Furthermore, the difference between the two independent parameters is multiplied by the mutation, i.e. for each individual member a mutant component is generated (Fan, 2019). In other words, it helps to generate optimal outcomes by making adjustments in the parameters and helps to track the status of the current search by storing the current knowledge in memory (Babu & Munawar, 2007). DE contains the strong capacity of global coverage that can help to get the best outcomes. On the other hand, GA and PSO have huge amount of parameters that enormously impact the outcomes. A large number of parameters actuates the difficulty level of the algorithm to optimise the eminent problems in real applications. Therefore, DE has a better capability of the parameter adjustment. The detailed procedure of DE-based ANFIS optimisation has been mentioned ahead.

1. In the first step of optimisation, the population, i.e. the instances of air pollutant in a given space-time window (Δt,Δs) are originated, which is utilised to provide the input to the proposed ANFIS model. The population is formulated in the form of matrix $N \times D$ as presented by SVD. The variable N represents the number of vectors and D represents the total number of instance. The process of population generation, i.e. data accumulation for the *ith* vector is represented as follows:

$$X_{i,j} = l[j] + rand * (h[j] - l[j]) \tag{6}$$

   in which, i and j represents the number of rows and column of the matrix. The rand variable represents the uniform probability distribution-based random number. The $h[j]$ is the upper bound and $l[j]$ is the lower bound of the *jth* column's, respectively.
2. After performing initialisation i.e. incorporation of temporal instances, the mutant vectors are generated by performing mutation operation.

$$V_i = X_{r_1} + F * (X_{r_2} - X_{r_3}) \tag{7}$$

   where V_i is the mutant vector related to each target value X_i for the current population. F represents the factor of mutation. The variable $r_n \in 1, 2, \ldots, N$ represents the random factor numbers and the value of these random variables are completely different from each other. Mathematically, the non-equality of these random numbers is represented as $r_1 \neq r_2 \neq r_3 \neq i$.
3. After performing the mutation operation, crossover is applied by DE. Basically, crossover methods are responsible for making diversity in the population. The trial vector U_i is generated by crossover procedure which is used to set in between X_i and V_i. The binomial operator is considered as the most commonly used operator in crossover operation. The crossover process for each component is represented as follows:

$$U_{i,j} = \begin{cases} V_{i,j}, & \text{if } (rand(j) \leq C \text{ or } j = j_{rand}) \\ X_{i,j}, & \text{otherwise} \end{cases}$$

   where C represents the crossover rate and j_{rand} represents the randomly chosen index within [1,D].

4. The next step of selection is employed to keep the size of the population by determining the target or trial vectors for the next population generation. The selection process based on the one-to-one tournament is represented as follows:

$$X_i = \begin{cases} U_i, & \text{if } f(U_i) \leq f(x_i) \\ X_i, & \text{otherwise} \end{cases}$$

where the objective function is represented by f(x) which needs to be optimised.

The methodical progression of the proposed ANFIS model with DE optimiser for parameter tuning is elaborated in Algorithm 2 (Table 3). The scale of different air quality parameters can be recognised by utilising the interdependence relation. The parameters of the DE algorithm are tabulated in Table 4. The current research summarises these mathematical optimisation steps as follows.

(1) In the initial step, the population for the initial input, i.e. spatio-temporal instances of air pollutant is generated and the parameters of DE optimiser are specified in the initial population.
(2) After the generation of the input population, the rules of FIS are designed by using membership functions. The variables used for input and output are further tuned by correlating the parameters of the DE.
(3) The error is calculated by performing the RMSE error function for each particle solution. By analysing the scale of error, the values are updated to optimise the prediction capability of the system.
(4) In the final step, the set of optimal parameters are obtained to predict air pollutant like benzene and its concentration over spatio-temporal domain more accurately.

Decision visualisation layer (DVL)

Decision Visualisation Layer (DVL) is used as a presentation layer to the users especially health-conscious patients for effective delivery healthcare services. The major task of this layer is to present the predicted air quality values to the user on a hand-held device. For feasibility, Liquid Crystal Display (LCD) screen is required for displaying the results to users in real-time depicting the air quality parameters in a given region over a given timescale. Moreover, instead of the numerical display of parameters, the visualisation of DVL is enhanced by Self-Organised Mapping (SOM) technique. SOM technique enables dynamic presentation based on colour-coding mechanism. Proposed ANFIS-DE model is visualised using U-matrix for implementing the SOM technique. This is augmented with bar graph based result generation for visualisation. There are many tools commercially available for SOM-based visualisations. The details of U-Matrix can be found in (Bhatia & Sood, 2017). Figure 6 depicts the utilisation of U-Matrix for SOM visualisation. The high value of m-AQI is denoted by red colour and low m-AQI is presented using green colour. In other

Table 3. ANFIS-DE Technique of m-AQI Prediction.

Algorithm 2: Air Pollutant Prediction Mechanism using ANFIS Technique with DE Optimisation
Input: Input training dataset
Output: Optimised set of FIS parameters for scale prediction of benzene in the air
Step 1: Generation of the initial population and specifying the initial parameters of the DE algorithm
Step 2: Specifying membership functions with fuzzy rules for designing initial FIS
Step 3: Specifying the total number of iterations at a particular time module ΔT.
Step 4: Tuning input and output variables using DE to get rule-based results.
Step 5: Calculating the range of fitness related to each particle
Step 6: Evaluating the loss rate by calculating RMSE for generating new population-based optimal values
Step 7: Updating all the values of the previous population
Step 8: Updating the value of epoch for next time module ΔT = t + 1, where t is the instance of a particular time T module
Step 9: Creating a fuzzy model by obtaining optimal parameters
Step 10: Return optimal parameters and Goto Step 4.

Table 4. Parameters of DE algorithm.

Dimensions	Size of the parameters
Size of Population	20
Mutation	0.5
Crossover	0.9
Boundaries	12

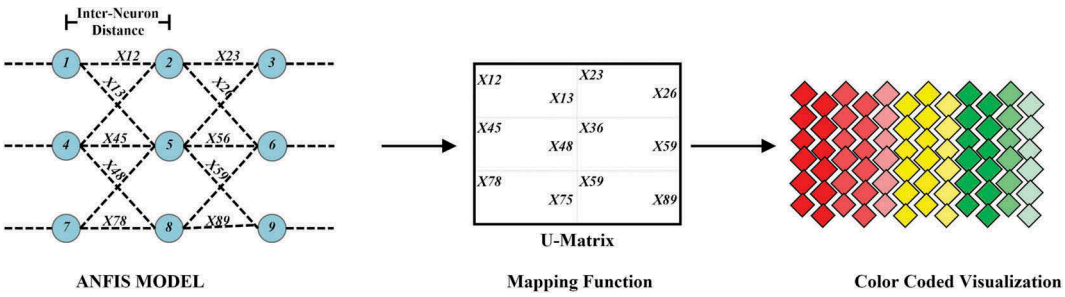

Figure 6. SOM-technique for Visualisation.

words, numerical values are scaled on the LCD screen with colour indication using bar graphs. This enables the user for a better indication of severe air quality condition so that necessary steps can be taken by users. Generally, it is depicted as follows.

High m-AQI Value → Red Colour
Medium m-AQI Value → Yellow Colour
Low m-AQI Value → Green Colour

It is important to mention that data transmission from fog computing device to LCD display can be performed using numerous wireless technologies like blue-tooth, WiFi, and ZigBee. In addition to the prediction of vulnerable pollutants, an alert generation procedure is added to provide enhanced functionality to the users suffering pollutant-related diseases. This can be significantly helpful especially for the patients suffering from diseases like asthma, and COPD. The entire procedure of visualisation with alert generation is presented in Table 5. An illustration of SOM-Technique for New Delhi region is shown in Figure 7.

Experimental implementation

The proposed IoT-Fog inspired Smart Air quality Prediction system is divided into three steps. Initially, data is acquired ubiquitously using numerous IoT sensors embedded in a geographically distributed

Table 5. Visualisation and Alert Generation.

Algorithm 3: Visualisation and Medical Alert Generation Procedure
Input: Input IoT dataset for Air Pollutants in Ambient Environment **Step 1**: Input the data values for air pollutants. **Step 2**: Determine current window (Δt,Δs) **Step 3**: Store IoT values in data repository (fog device) for analysis. **Step 4**: Compute m-AQI using *Algorithm 1*. **Step 5**: Begin prediction procure using *Algorithm 2*. **Step 6**: If *m-AQI > Threshold (pre-defined by domain expert)*, Generate Medical Alert Signal **Step 7**: Present values to user SOM techniques. **Step 8**: Stop

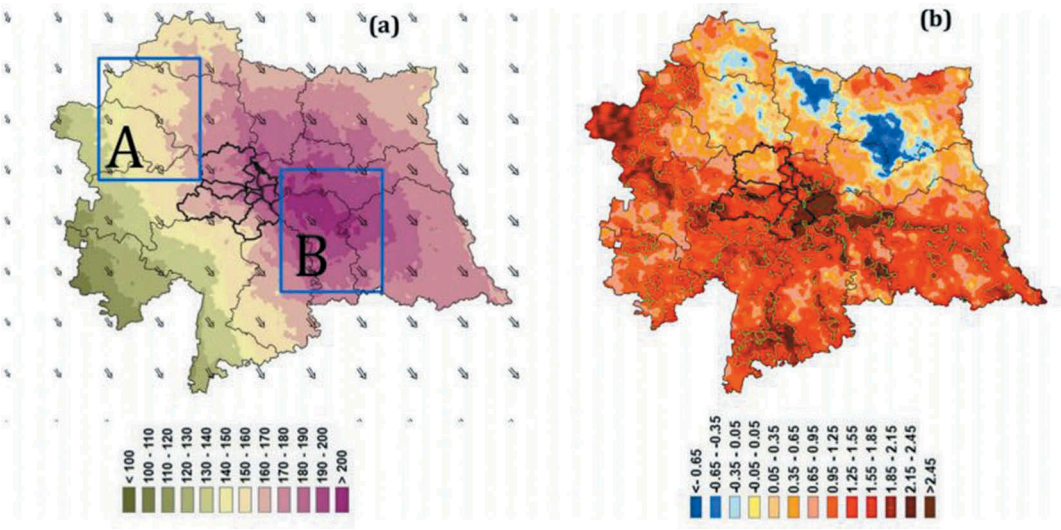

Figure 7. SOM-Instance in New Delhi, India Region for PM(2.5) Concentration; (a) Mean ambient PM2.5 ground-level concentrations 29 October 2018–4 November2018; upwind (A) and downwind (B), (b) 31 December 2019–6 January 2019 (Chowdhury et al., 2019).

region. This data is analysed using a proposed spatio-temporal mining technique. Finally, machine learning is used to predict the air quality in terms of the Level of Pollutant (LoP) using ANFIS-DE technique. Henceforth, the experimental simulation is performed to realise the following objectives.

(1) Determine the temporal efficiency of the proposed model for real-time sensitiveness.
(2) Statistically identify the performance parameters in terms of Accuracy, Coefficient of Determination, Root Mean Square Error for the proposed machine learning-based ANFIS model.
(3) Identify the reliability aspect of the presented technique of air quality prediction.
(4) Evaluate the overall system stability in terms of Average Absolute Shift (AAS) for efficiency assessment over temporal basis.

Simulation environment

Simulation Environment adopted for experimental implementation is inspired by real-world application. Real-time air quality monitoring and LoP prediction have been considered to determine the overall performance of the system. Four challenging datasets of air quality parameters have been acquired from the online data repository of UCI and Kaggle. These include Air Quality Dataset of Italian City (AQD-IC) with nearly 9358 instances,[5] Pollution Data from Refinery Monitoring (AQD-RM) with 31643 data instances,[6] Air Quality Dataset of Gijon City (AQD-GC) with 72276 instances[7] and Air Quality Data set for Northern Taiwan (AQD-NT) with 218641 instances.[8] These datasets are analysed using Raspberry Pi V3 device (fog computing node) having the capacity of 16GB and 1.2 GHz clock cycle, which is further connected to Amazon EC2 cloud for data storage. Figure 8 shows an overview of the Raspberry Pi device used for the current scenario with various specifications. The simulation was performed using MATLAB toolkit for result generation. For comparative analysis, numerous state-of-the-art models are considered to determine the performance enhancement of the proposed system. In addition to this, several statistical parameters are evaluated to determine the overall efficacy of the proposed system.

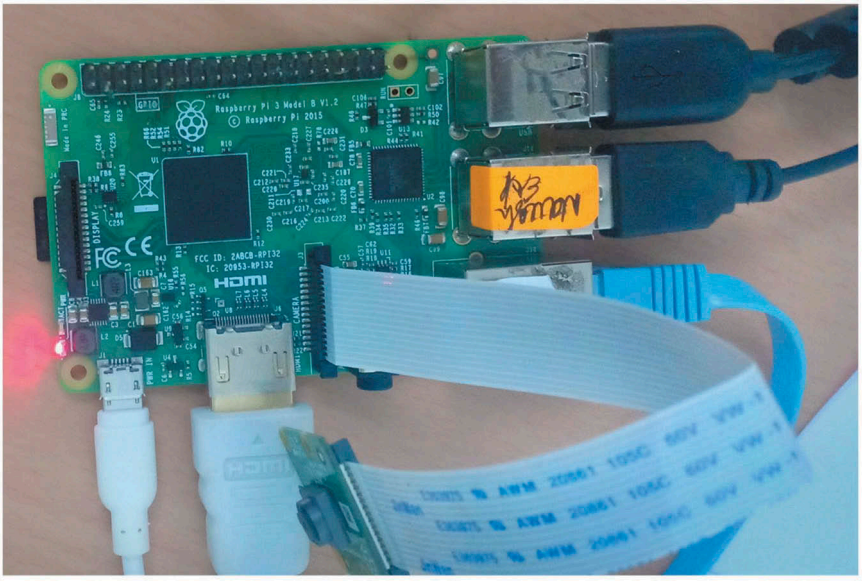

Figure 8. (a) Raspberry Pi V3; Specification: Quad-core 64-bit ARM processor with 1.2 GHz clock cycle; RAM memory is nearly 1 GB with VideoCore IV GPU graphical capability; 16GB of Internal Storage.

Temporal efficiency

Temporal efficiency is concerned with the time taken to generate the prediction results for the proposed model. In other words, temporal efficiency enables feasible evaluation of temporal delay from the time of data insertion and to the generation of output results. Mathematically, Temporal Delay is evaluated as follows.

$$Temporal\ Delay\ \Delta t = Time_{Prediction} - Time_{Allocation}$$

where $Time_{Prediction}$ denotes the time instance for generation of results and $Time_{Allocation}$ denotes the time at which data was fed to the system. For determining the temporal efficiency of the proposed model, results are compared with several state-of-the-art prediction models, namely, K-Nearest Neighbour (K-NNR), Decision Tree Regression (DTR), and Support Vector Regressor (SVR). Detailed simulation results are discussed ahead.

(1) The plot in Figure 9(a) shows execution time for the proposed model using AQD-IC datasets. Nearly 9358 instances were used to analyse the running time for the proposed model. For predictability results, the presented model registered temporal delay averaging to 15.36 s. In comparison to this, SVR took an average running time of 18.39 s. Similarly, using K-NNR as prediction model, the overall execution time was averaged to 23.78 s. Similarly, the running time for DTR datasets was approximated to 25.69 s for AQD-IC datasets. Based on these results, it is concluded that the presented model is temporally efficient in the current scenario as compared to other prediction models.
(2) The plot in Figure 9(b) depicts the execution time of the presented model using AQD-RM datasets comprising nearly 31,643 data instances. Based on the experimental simulation, the proposed model was able to acquire temporal delay numerating to 25.98 s. Comparatively, SVR was able to register high temporal delay of 29.78 s averaging over the number of instances. Similarly, K-NNR and DTR were able to perform prediction with the average running time of 33.48 s and 38.12 s, respectively.

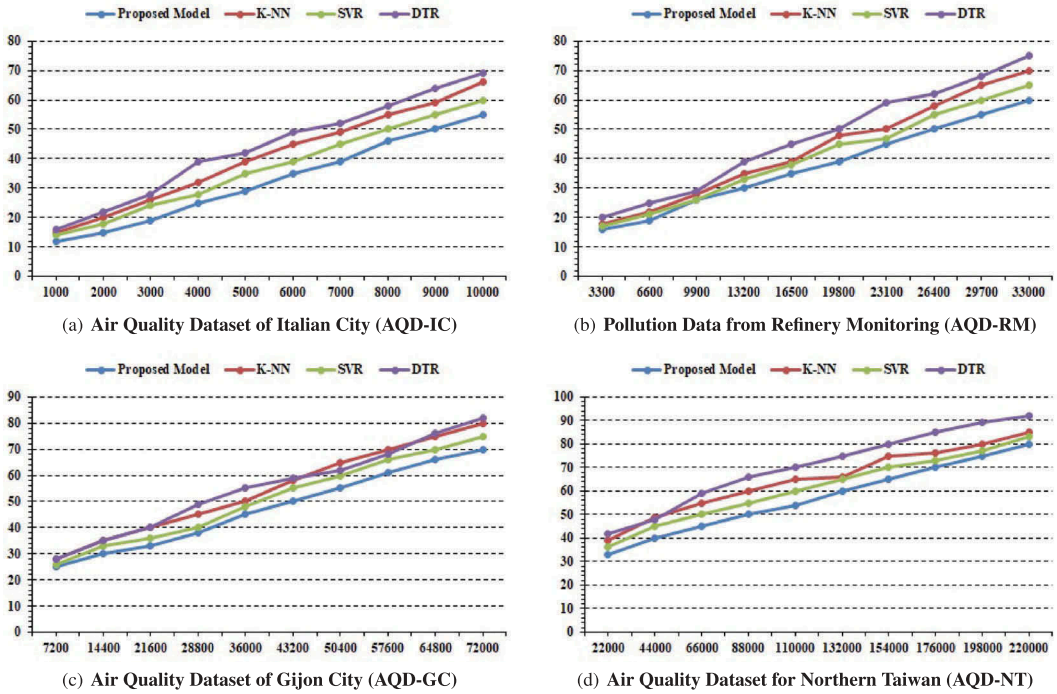

Figure 9. Temporal Efficiency Comparison.

(3) The plot in Figure 9(c) shows the temporal delay using AQD-GC dataset comprising 72,276 environmental instances. Based on the experimental simulation, the presented model was able to acquire temporal delay averaging to 30.01 s, which is better than 36.69 s of SVR, 40.11 s of K-NNR and 44.12 s of DTR. Henceforth, it can be concluded that the presented model is able to outperform other prediction models effectively.

(4) The plot in Figure 9(d) presents the results of temporal delay when simulated using AQD-NT with 218,641 data instances. In the current scenario, the proposed system is able to register average temporal delay of 35.69 s which is better in comparison to SVR (39.45 s), K-NNR (42.98 s) and DTR (45.88 s). Based on these results, it is concluded that the proposed model is temporally better in comparison to other state-of-the-art prediction models for efficient prediction of air pollutants.

Prediction efficiency

Table 6–8 depicts the calculated results for the determination of prediction efficiency by the proposed methodology based on different datasets. Specifically, it shows the relationship between the parameters in terms of Coefficient of Determination (*r2*), and Root Mean Square Error (RMSE) values registered during testing on the datasets. For comparative analysis, numerous other prediction models were incorporated in addition to K-NNR, SVR and DTR. These prediction techniques include Artificial Neural Network (ANN), Stochastic Gradient Descent with Regressor (SGDR). Moreover, numerous ensemble techniques like Random Forest Regressor (RFR), Adaptive Boosting (AdaBOOST), Gradient Boosted Decision Trees (GBDT) and ANFIS were used for comparative analysis. However, it is important to mention that during implementation only the prediction model is altered while the rest of the model is kept identical. Detailed results are mentioned ahead.

Table 6. Results for Accuracy.

Techniques	Methods	AQD-IC	AQD-RM	AQD-GC	AQD-NT
Base-line Techniques	k-NNR	88.1	88.8	89.1	90.6
	SVR	88.9	88.5	89.6	92.7
	DTR	89.6	90.7	92.1	93.3
	ANN	90.4	91.4	93.2	94.5
	SGDR	91.9	92.1	94.7	95.1
Ensemble Technique	RFR	92.7	93.2	95.1	95.9
	ADABOOST	94.7	94.7	96.8	96.4
	GBDT	96.1	96.1	97.2	95.2
	ANFIS(without DE)	95.1	96.4	97.6	97.1
	PROPOSED (ANFIS-DE)	**97.3**	**97.6**	**98.9**	**98.9**

Table 7. Results for Coefficient of Determination (*r2*).

Techniques	Methods	AQD-IC	AQD-RM	AQD-GC	AQD-NT
Base-line Techniques	k-NNR	0.77	0.82	0.79	0.77
	SVR	0.78	0.84	0.80	0.79
	DTR	0.76	0.83	0.77	0.76
	ANN	0.80	0.85	0.80	0.78
	SGDR	0.81	0.87	0.82	0.80
Ensemble Technique	RFR	0.83	0.89	0.85	0.82
	ADABOOST	0.85	0.90	0.87	0.83
	GBDT	0.87	0.91	0.89	0.85
	ANFIS(without DE)	0.85	0.93	0.91	0.89
	PROPOSED (ANFIS-DE)	**0.94**	**0.95**	**0.93**	**0.93**

Table 8. Results for Root Mean Square Error (RMSE).

Techniques	Methods	AQD-IC	AQD-RM	AQD-GC	AQD-NT
Base-line Techniques	k-NNR	4.2	5.9	3.9	3.2
	SVR	3.6	4.1	3.4	3.6
	DTR	5.2	5.3	2.8	4.1
	ANN	5.7	4.5	3.2	4.6
	SGDR	4.1	4.8	2.7	2.8
Ensemble Technique	RFR	3.8	5.2	2.9	2.5
	ADABOOST	4.6	3.7	2.6	1.9
	GBDT	4.9	3.9	2.3	2.3
	ANFIS(without DE)	3.9	2.8	2.1	2.6
	PROPOSED (ANFIS-DE)	**2.4**	**1.9**	**1.0**	**1.7**

(1) Table 6 demonstrates the accuracy value for the proposed prediction technique. Its shows that the presented methodology has achieved higher accuracy numerating to 97.3% as compared to the other baseline and ensemble techniques in the prediction phase with minimal variations on the AQD-IC dataset. Similarly, for AQD-RM dataset, the proposed model was able to register the accuracy value of 97.6% which is comparatively better than other techniques. Moreover, for AQD-GC and AQD-NT dataset, the presented technique was able to outperform other techniques and registered accuracy of 98.9% and 98.9%, respectively. Based on these results it can be concluded that the presented technique of ANFIS-DE for prediction is highly accurate in the current scenario.

(2) Table 7 shows another important statistical parameter named as the Coefficient of Determination (COD) (r2). Mathematically,

$$r^2 = 1 - \frac{SSE}{SST}$$

where total sum of squares $SST = S_{yy}$ and sum of squared residuals $SSE = S_{yy} - S_{xy}^2/S_{xx}$. The

Table 9. Comparative Analysis of Air Quality Models.

Parameters	Pannu et al. (2018)	Perez et al. (2019)	Caron et al. (2019)	Nihalani et al. (2020)	Balram et al. (2019)	Pannu et al. (2017)	Proposed Model
Applicability Domain	Air Quality Prediction Model	Remote Air Quality Monitoring	Air Quality Treatment Procedure	Air Quality Assessment	Air Quality Monitoring	Air Quality Monitoring	IoT-Fog based Air Quality Monitoring System
Major Contribution	Predictive model for air quality exposure	IBM Computer System Usability Questionnaire based Healthcare Model	Utilisation of K-means algorithm to analyse data acquired by online gas analyser	Quantify air pollutant concentrations for four cities in Gujarat from 2011 to 2015	Air quality warning system	Predictive Air Quality Model	Analysis of air quality using IoT sensors and prediction of Air Quality Index in real-time
Baseline Sensing Technology Used	None	None	Gas Sensors, and Miniatures	None	None	None	IoT
Data Classification Model used	None	Generalised Air Quality	K-Means Algorithm	None	Support Vector Machine	None	Bayesian Belief Network
Numerical Quantification	Yes	No	No	Yes	Yes	Yes	Yes
Feature Extraction Mechanism	Yes	Not Applicable	Yes (Clustering)	Not Applicable	None	Yes	Heterogeneous
Data Mining Technique	Not Applicable	Not Applicable	Multiple	Multiple	Multiple	Not Applicable	Spatio-Temporal Data Mining
Data Storage Repository	Local	Local	Local	Local	Statistic	Local	Local and Cloud
Output Presentation	No	No	No	No	No	No	Yes (SOM)
Decision Making Model	ANFIS-PSO	Alert Based	Not Applicable	Fuzzy Logic	Bayesian Regularised Neural Network (BRNN)	ANFIS-PSO	ANFIS-DE
Heterogenous Dataset	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Security Mechanism Implied	No	No	Yes	No	No	No	Yes
Time Complexity	NA	NA	NA	O(n2.5)	O(n2log(n))	NA	O(n2)

values of *Sxx, Sxy, Syy* are the values obtained by testing and prediction, which are interpolated over the x-y plane. It can be seen that the presented technique was able to outperform other techniques in evaluating COD measure. Specifically, for AQD-IC dataset, the presented technique registered COD value numerating to 94% which is far better as compared to other prediction techniques. Similarly, for AQD-RM, AQD-GC, and AQD-NT, the COD value for the proposed technique was nearly 95%, 93%, and 93%, respectively, which is much than comparative models. Based on these results, it can be concluded that the proposed system attains better performance and is highly suitable for the current scenario of air quality determination.

(3) Table 8 depicts the results for Root Mean Square Error (RMSE). RMSE is an important measure for determination of error in the prediction values for the proposed model. Mathematically.

$$RMSE = \sqrt{\frac{1}{n}\Sigma_{i=1}^{n}\left(\frac{x_i - y_i}{\sigma_i}\right)^2}$$

where σi denotes the error obtained from training of the model for ith instance and *n* denotes the number of instances. It can be seen that the proposed methodology achieved less RMSE (2.4%) for AQD-IC which further shows that the prediction strength of the model. Moreover, for AQD-RM, AQD-GC, and AQD-NT, the RMSE was registered with minimal values (1.9%, 1.0%, and 1.7%, respectively) as compared to other baseline and ensemble techniques. Henceforth, it can be concluded that the presented technique is comparatively better and accurate for the prediction of air pollutants in real-time.

Reliability analysis

For the purpose of estimating the reliability of the proposed system, only the prediction model is altered while the remaining model is identical. Numerous simulations are performed on the proposed system and results are depicted in Figure 10. The reliability trends for different prediction models determine the overall system accuracy. As the number of datasets are increasing for prediction purpose, the proposed ANFIS-DE model is able to outperform other models. As it can be seen that, in the current scenario the presented technique is able to register average reliability of 92.36% as compared to 91.36% of K-NN, 90.26% of SVR and 91.105 of DTR. Moreover, the consistently high value of reliability depicts the enhancement in prediction accuracy by the proposed technique.

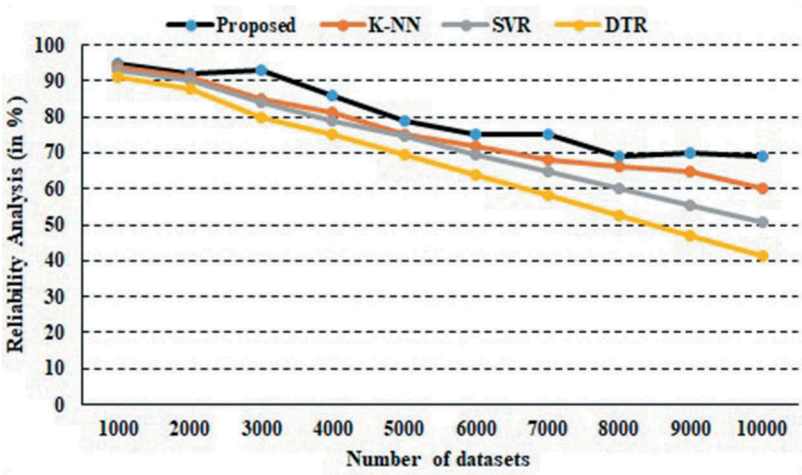

Figure 10. Reliability Analysis.

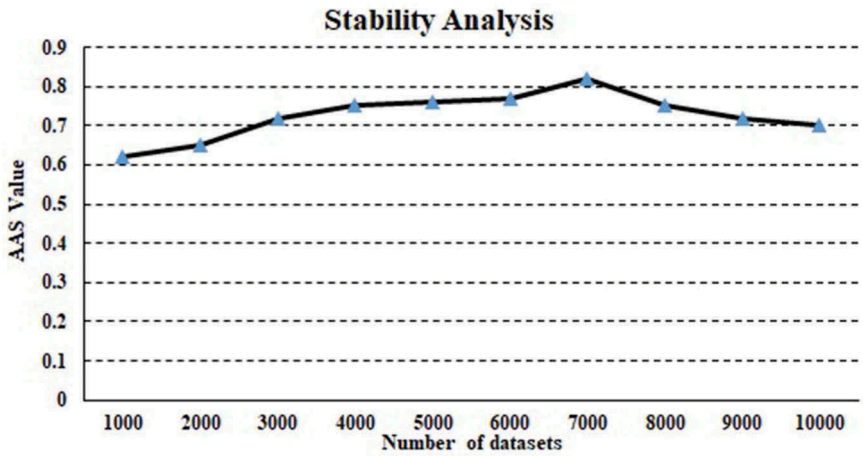

Figure 11. Stability Analysis.

Henceforth, it can be concluded that in the current scenario, the proposed technique of air quality prediction is more reliable than other state-of-the-art prediction models.

Stability analysis

System stability is a prevalence measure to determine the overall performance of a system. In other words, it is used to estimate the overall stabilisation over a large number of datasets. Specifically, system stability is measured in terms of Average Absolute Shift (AAS). The value of AAS measure lies between (`) where value 0 denotes minimum stability while value 1 denotes maximum stability. Figure 11 shows the results acquired for the proposed model for stability estimation. It can be depicted that the presented technique of air quality prediction obtains a minimum value of 0.62 and maximum value 0.82 for AAS, averaging to nearly 0.76. Based on these results, it is concluded that the proposed system is highly stable for analysing large datasets for air quality monitoring and healthcare provisioning.

Comparative analysis

This section provides a comparative analysis between the presented technique of air quality monitoring system and other related studies. As mentioned earlier, machine learning techniques have been developed to analyse large data in real-time. Inspired from these aspects, there are numerous distinguishing features in comparison to other state-of-the-art technique. Specifically, five challenging studies have considered for comparative analysis. These include studies performed in Caron et al. (2019), Balram et al. (2019), Nihalani et al. (2020), Pannu et al. (2018), (2017), and Pérez-Medina et al. (2019). Numerous parameters are identified for comparative analysis as discussed ahead. A summary of the comparison is provided in Table 9.

1. *Applicability Domain*
   This parameter provides the domain of application for the proposed model. In other words, it depicts the overview of applicability for which the presented technique has been developed.
2. *Major Contribution*
   This parameter provides in-depth information about the major contribution and objectives of the presented system. Moreover, it provides an influential aspect of specific research from the industrial perspective.

3. *Baseline Sensing Technology Used*
   It provides information about the way in which data is acquired for air quality monitoring in real-time. Moreover, as accurate prediction is one of the objectives of the current study, it becomes indispensable to determine the underneath data sensing techniques utilised in the presented system.
4. *Data Classification Model Used*
   As discussed in the previous sections about the incorporation of air quality data for classification in harmful datasets, this parameter will provide the technique used for classification of air quality parameters in real-time by different studies for vulnerability determination.
5. *Numerical Quantification*
   Numerical quantification of air quality data is another novel concept of the presented research. However, it is necessary to compare different studies for the quantification of air quality parameters in real-time.
6. *Feature Extraction Mechanism*
   Air quality parameters comprise heterogeneous data elements which require effective feature extraction mechanism for the overall system. This parameter compares the presented feature extraction technique that has been incorporated in the current research with respect to other related studies.
7. *Data Mining Technique*
   Real-time air quality monitoring and prediction require efficient data mining technique for analysing the complex heterogeneous data. For this purpose, the data mining technique determines the abstraction of data for effective numerical analysis.
8. *Data Storage*
   Data storage presents the storage mechanism utilised for accumulating real-time air quality data. Based on the data repository, further analysis can be performed effectively. Henceforth, this parameter provides the data storage technique used in different studies.
9. *Output Presentation*
   This parameter provides information about the way in which air quality is presented to the user. Since data analysis is performed at the back-end, it becomes indispensable to provide information about air quality to the user in real-time effectively.
10. *Decision Making Model*
    It is an important parameter for determination of the decision-making model that has been incorporated in the presented study. In other words, it provides information about the way in which the prediction of air quality has been performed in the presented research.
11. *Heterogeneous Datasets*
    This parameter provides information about the type of data that has been incorporated in the presented research. In other words, it defines the type of data segments utilised for analysis of air quality for providing visualisation services.
12. *Security Mechanism Implied*
    Data security is one of the major aspects of air quality data. As any alteration of air quality can result in severe effects. Henceforth, data security technique is necessary for the protection of air quality data at the network layer and storage layer.
13. *Time Complexity*
    Time complexity provides the running time taken by the proposed algorithm in the presented algorithm.

Conclusion

Air pollution has been considered as one of the major causes of health diseases around the world. In addition to consequencing numerous respiratory diseases, harmful production of air pollutant such

as benzene, sulphur dioxide has resulted in the deaths of a large number of people around the world. In this paper, a comprehensive machine learning inspired air pollutant monitoring and prediction system for smart air quality management has been presented to provide ubiquitous healthcare services. Specifically, IoT devices are used for data acquisition in the ambient environment, which is analysed using ANFIS technique for prediction of air quality parameters such as benzene in terms of LoP value. Specifically, IoT devices are used for data acquisition in the ambient environment, which is analysed using modified ANFIS-DE technique for prediction of air quality parameters in terms of m-AQI value. For validation purpose, numerous simulations have been performed and results are compared with state-of-the-art prediction models. Based on simulations, it is concluded that the proposed system is able to outperform other models in terms of various statistical parameters (Accuracy, Coefficient of Determination and Root Mean Square Error). In addition to this, the proposed system is analysed for reliability and stability to depict enhanced performance.

There are numerous open challenges that must be encountered for carrying out future research. Network efficiency is one of the issues for IoT service provisioning. Moreover, human-computer interaction for visualisation is another important domain for further exploration in the current domain.

Notes

1. Source: https://www.download.thelancet.com/gbd.
2. https://www.who.int/en/news-room/actualitysheets/detail/encompassing-outside-air-quality-and-wellbeing.
3. https://www.lung.org/about-us/media/official statements/2018-condition of-the-air.html.
4. Source: https://ourworldindata.org/air-pollution.
5. Source: https://archive.ics.uci.edu/ml/datasets/Air+quality.
6. Source: http://www.fenceline.org/richmond/data.php.
7. Source: https://transparencia.gijon.es/.
8. Source: https://taqm.epa.gov.tw/taqm/tw/YearlyDataDownload.aspx.

Disclosure Statement

The authors declare that they have no conflict of interest with anyone.

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